# Overconfidence in Confidence Interval Estimates Produced by Frontier Large Language Models

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**We document systematic miscalibration in confidence interval estimates produced by three frontier large language models (Claude, GPT-4, Gemini). In Study 1, all three models were substantially overconfident when predicting 1-day equity prices, with GPT-4 exhibiting the largest Expected Calibration Error (ECE = 43.7%) and Claude the smallest (ECE = 19.7%). In Study 2, using 100 equities across nine prediction horizons spanning 1–22 days ($N > 18{,}000$ scored predictions), Claude and GPT-4 became increasingly miscalibrated at longer horizons, whereas Gemini maintained near-target coverage at the longest horizon tested (ECE = 2.5% at 22 days). In Study 3, extending to six heterogeneous prediction domains, we found a systematic domain-confidence inversion: all models were overconfident in commodity markets ($\mu = 0.25$ uniformly) but underconfident in cryptocurrency ($\mu = 1.35$–$14.4$), while achieving near-target calibration on weather. Across all three studies, GPT-4 was the most overconfident model and Claude the most calibrated. These findings show that LLM miscalibration is domain- and horizon-structured rather than uniform, with direct implications for deploying LLMs in high-stakes decision support.**

large language models | calibration | overconfidence | interval estimation | forecasting

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## 1. Introduction

A well-calibrated forecaster states 80% confidence intervals that contain the true outcome 80% of the time. Decades of research on human judgment document systematic failure to achieve this standard: people's intervals are too narrow, a phenomenon termed overconfidence in interval estimates (1–4). The consequences are severe. Overconfident interval estimates underlie miscalibrated risk assessments in medicine (5), finance (6), engineering (7), and strategic planning (8).

Large language models (LLMs) are now routinely used as decision-support tools in precisely these domains, yet whether their probabilistic forecasts are calibrated remains an open question (9–11). Prior work has focused on factual question-answering confidence or binary classification probabilities. Virtually no study has examined LLM calibration on interval estimates for real-world quantitative forecasting, or asked how miscalibration varies across prediction horizons and domains.

We address these questions with three preregistered studies. Study 1 establishes a baseline: are three frontier LLMs overconfident when predicting next-day equity prices with stated confidence intervals? Study 2 extends to nine prediction horizons (1–22 days) and asks: does miscalibration scale with forecast horizon? Study 3 extends across six heterogeneous domains (equities, commodities, cryptocurrency, foreign exchange, weather, NBA game totals) and asks: is miscalibration domain-specific, and can its direction reverse?

The human overconfidence literature consistently shows that miscalibration is domain-specific: experts are most overconfident in their own fields of ostensible competence (3, 4). It also shows that overconfidence in interval estimates increases with task difficulty and time horizon (12). If LLMs inherit domain and horizon structure from training data, they may replicate or amplify these asymmetries.

Our results confirm and extend the human literature. GPT-4 is the most overconfident model in every study; Claude is the most calibrated. Study 2 reveals a model divergence at longer horizons: Claude and GPT-4 become increasingly miscalibrated as the prediction window grows, while Gemini maintains near-target coverage even at 22 days. Study 3 uncovers a domain-confidence inversion—all models overconfident in structured financial markets (equities, commodities) but underconfident in speculative markets (cryptocurrency, forex)—a pattern absent from prior accounts of LLM calibration. This paper contributes to the LLM calibration literature (9, 11) by focusing on interval estimates rather than point probabilities and by using real-world outcomes rather than knowledge-base queries.

**Study 1.** Study 1 established a baseline calibration estimate for three frontier LLMs producing confidence intervals for 1-day equity price predictions. We predicted that all models would be overconfident—stating intervals narrower than warranted by realized outcomes—and that the degree of miscalibration would differ systematically across models.

***Methods.*** Three commercially deployed LLMs were evaluated: Claude (claude-sonnet-4-20250514; Anthropic), GPT-4 (gpt-4o; OpenAI), and Gemini (gemini-2.5-flash; Google DeepMind). All models were queried via their public APIs at default temperature with a maximum of 512 output tokens per call.

**Stimuli and predictions** Twenty large-cap S&P 500 equities were selected as prediction targets (AAPL, NVDA, MSFT, AMZN, GOOGL, META, AVGO, TSLA, BRK-B, WMT, LLY, JPM, XOM, V, JNJ, MU, MA, ORCL, COST, SPY). Each prompt provided the current closing price and prediction horizon as plain text, requesting a point estimate plus 50%, 80%, and 90%

confidence intervals as structured JSON. No chain-of-thought reasoning or calibration instructions were provided; models were evaluated in their default, deployment-ready configuration. Predictions were collected on March 23, 2026 for outcomes on March 24, 2026. Each model made 5 independent runs per ticker, yielding 100 scored predictions per model.

**Calibration metrics** For each confidence level $\alpha \in \{0.50, 0.80, 0.90\}$, the empirical *hit rate* $\hat{p}_\alpha$ is the proportion of predictions for which the true outcome fell within the stated interval $[L, U]$. A calibrated model achieves $\hat{p}_\alpha = \alpha$; values below indicate overconfidence. Expected Calibration Error (ECE) is the mean absolute deviation between stated and empirical coverage across all three levels:

$$\mathrm{ECE} = \frac{1}{3} \sum_{\alpha \in \{0.50,\, 0.80,\, 0.90\}} |\alpha - \hat{p}_\alpha| \,. \qquad [1]$$

A perfectly calibrated model achieves $\mathrm{ECE} = 0$.

***Results.*** Table 1 summarizes calibration for all three models.

**Hit rates** All models were substantially overconfident at every confidence level. Claude was the least miscalibrated (Hit@80 = 58%, Hit@90 = 73%), but remained far below calibration targets. GPT-4's 80% CI contained the true outcome only 31% of the time, and its 90% CI only 38%—implying interval widths roughly consistent with a 30% confidence level. Gemini closely tracked Claude at the 80% level but underperformed at 90% (65%).

**ECE** GPT-4's overall ECE (43.7%) was more than double Claude's (19.7%) and more than triple the perfect-calibration target of 0%. Gemini was intermediate (22.7%).

**Table 1. Study 1: Calibration in 1-day equity predictions ($n = 100$ per model). ECE = Expected Calibration Error; lower is better.**

| Model | Hit@50% | Hit@80% | Hit@90% | ECE |
|---|---|---|---|---|
| Claude | 30.0% | 58.0% | 73.0% | 19.7% |
| Gemini | 29.0% | 58.0% | 65.0% | 22.7% |
| GPT-4 | 20.0% | 31.0% | 38.0% | 43.7% |
| Perfect | 50.0% | 80.0% | 90.0% | 0.0% |

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***Discussion.*** All three models produced systematically narrow confidence intervals for a simple, single-domain forecasting task. GPT-4's calibration (31% coverage at the 80% level) is comparable to the most severe human expert overconfidence on record: **(author?)** (6) found that CFOs' 80% intervals for annual equity returns achieved only 36–38% coverage. These results confirm universal overconfidence in the simplest possible setting and establish the model ranking—Claude best, GPT-4 worst, Gemini intermediate—that will prove consistent across Studies 2 and 3 in the aggregate.

**Study 2.** Study 1 established overconfidence at a single, short horizon. Study 2 asked whether stated confidence intervals—which must widen to maintain coverage as forecast uncertainty grows—actually scale with prediction horizon. Using a backtesting design with 100 large-cap equities and nine horizons spanning 1 to 22 trading days, we tested whether miscalibration grows, shrinks, or diverges across models as the horizon extends.

***Methods.*** The same three models were evaluated using the same prompt format and structured JSON output as Study 1. Study 2 used a backtesting design in which predictions were made against historical prices using date-blinded prompts (current price and horizon provided; calendar date withheld) to prevent models from conditioning on known future events.

**Design** One hundred large-cap S&P 500 equities were selected. Predictions were collected at nine horizons: 1, 2, 3, 6, 7, 18, 20, 21, and 22 calendar days. Five independent predictions per ticker per model were collected at each horizon, using a reference date of March 24, 2026. Market closures were handled by applying a ±6-calendar-day tolerance window when matching predictions to realized prices. After excluding unresolved predictions, the design yielded between 157 and 1,495 scored predictions per model-horizon cell ($N > 18{,}000$ total).

**Calibration metrics** Hit rates and ECE were computed as in Study 1, aggregated within each model-horizon cell.

***Results.*** Table 2 reports 80% hit rates and ECE at four representative horizons; complete results for all nine horizons are in the Web Appendix.

**Horizon degradation: Claude and GPT-4** Both models exhibited marked degradation with increasing horizon. Claude's Hit@80 fell from 80.5% at 1 day to 46.2% at 18 days (ECE rising from 1.4% to 29.8%), consistent with CI widths anchored to short-horizon volatility that fail to scale with realized uncertainty at longer windows. GPT-4—already poorly calibrated at 1 day (ECE = 21.8%)—deteriorated further, reaching Hit@80 = 27.2% and ECE = 47.4% at 18 days. By 22 days, GPT-4's ECE (49.9%) approached the theoretical maximum for three-level calibration.

**Horizon robustness: Gemini** Gemini exhibited a qualitatively different pattern. Its 1-day calibration (ECE = 1.8%, Hit@80 = 78.5%) was the best of any model at any single horizon in the study. While it degraded at intermediate horizons (ECE = 16.6% at 18 days), it recovered strongly at 22 days (Hit@80 = 80.1%, ECE = 2.5%)—near-perfect calibration at the longest horizon tested.

**Table 2. Study 2: Hit@80 and ECE by model and prediction horizon (selected horizons; full table in Web Appendix).**

| | Claude | | Gemini | | GPT-4 | |
|---|---|---|---|---|---|---|
| Horizon | H@80 | ECE | H@80 | ECE | H@80 | ECE |
| 1d | 80.5% | 1.4% | 78.5% | 1.8% | 55.2% | 21.8% |
| 6d | 56.0% | 20.6% | 71.7% | 8.4% | 34.5% | 41.6% |
| 18d | 46.2% | 29.8% | 59.8% | 16.6% | 27.2% | 47.4% |
| 22d | 51.2% | 26.9% | 80.1% | 2.5% | 24.4% | 49.9% |

***Discussion.*** Prediction horizon strongly moderated miscalibration in a model-specific pattern. The degradation of Claude and GPT-4 mirrors the human tendency toward greater interval underestimation at longer horizons (12), consistent with CI widths anchored to short-run volatility. Gemini's recovery at 22 days is more surprising: its CI widths appear to scale approximately proportionally to realized volatility at the extremes of the tested range but not at intermediate horizons. Whether this reflects a genuine uncertainty representation or an incidental match with equity return volatility at these specific windows requires replication at additional horizons and across different historical periods.

**Study 3.** Studies 1 and 2 established overconfidence within a single domain (equities) and its moderation by horizon. Study 3 asked whether miscalibration is domain-general or structured by the type of quantity being predicted. Six heterogeneous domains—spanning financial markets, physical measurements, and sports outcomes—were included to test whether all models are uniformly overconfident or whether the direction and magnitude of miscalibration varies systematically with domain.

***Methods.*** The same prompt format and three confidence levels were used. Note that Study 3 employed a different Claude model version (claude-sonnet-4-6) than Studies 1 and 2 (claude-sonnet-4-20250514); GPT-4 (gpt-4o) and Gemini (gemini-2.5-flash) were identical across all studies (see Web Appendix for model version details). All predictions used a 24-hour horizon. Predictions were collected on March 25, 2026; outcomes were collected automatically on March 26, 2026. One prediction per item per model was collected.

**Domains and stimuli** Six domains were included: equities ($n = 20$ large-cap tickers, same set as Study 1), commodity futures ($n = 5$: gold, crude oil, silver, natural gas, copper), cryptocurrency ($n = 10$ by market cap), foreign exchange ($n = 8$ major pairs), weather ($n = 30$ major U.S. cities, next-day high temperature in °F), and NBA game totals ($n = 3$, subsequently excluded; see Web Appendix). Each domain prompt provided the current value as context and a question specifying the target quantity and date. Weather prompts additionally included the official meteorological forecast for the target day.* CoinGecko rate limits caused five cryptocurrency outcomes to remain unresolved; these were excluded. The final scored dataset comprised $n = 68$ resolved predictions per model ($N = 204$ total).

**Calibration and meta-knowledge metrics** Hit rates and ECE were computed as in Studies 1 and 2. We additionally computed the Soll–Klayman meta-knowledge ratio $\mu = \text{MEAD}/\text{MAD}$ (4), which quantifies whether stated uncertainty appropriately tracks actual forecast error. Treating 80% CI endpoints as the 10th and 90th percentiles of a symmetric normal ($z = 1.2816$), the Absolute Deviation for prediction $i$ is:

$$\text{AD}_i = \frac{|(L_i + U_i)/2 - y_i|}{\sigma}, \qquad [2]$$

where $\sigma$ is the within-domain outcome standard deviation. The meta-knowledge ratio $\mu = \text{MEAD}/\text{MAD}$ is the ratio of CI-width-implied error (MEAD) to actual midpoint error (MAD) at the domain level; $\mu < 1$ indicates overconfidence, $\mu = 1$ is perfect, and $\mu > 1$ indicates underconfidence.

***Results.***

**Overall calibration** Aggregating across all 68 resolved items, Claude was the best calibrated model (ECE = 5.3%), followed by Gemini (ECE = 7.2%) and GPT-4 (ECE = 18.9%; Table 3). Overall ECE values are lower than in Studies 1 and 2, driven by near-perfect calibration on weather—which accounted for 44% of Study 3 observations.

**Domain effects** Table 4 disaggregates calibration by domain. The central result is a systematic directional reversal: all models were overconfident in commodity and equity markets but underconfident in cryptocurrency and, to a lesser degree, forex. No model was uniformly overconfident or underconfident across all domains. Commodities showed the most extreme and uniform overconfidence (ECE = 40% for all three models; 80% and 90% CIs achieving only 40% coverage). Crypto showed the opposite pattern: Claude and GPT-4 both achieved 100% coverage at the 90% level. GPT-4's forex calibration was dramatically worse than the other models (ECE = 44.2%).

*Weather prompts provided the Open-Meteo forecast high temperature for the target date. Models produced CIs around (or independent of) this forecast. Full prompt templates are in the Web Appendix.

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**Table 3. Study 3: Overall calibration by model ($n = 68$ resolved predictions per model).**

| Model | Hit@50% | Hit@80% | Hit@90% | ECE |
|---|---|---|---|---|
| Claude | 58.8% | 82.4% | 85.3% | 5.3% |
| Gemini | 45.6% | 73.5% | 79.4% | 7.2% |
| GPT-4 | 30.9% | 63.2% | 69.1% | 18.9% |
| Perfect | 50.0% | 80.0% | 90.0% | 0.0% |

**Table 4. Study 3: Hit rates and ECE by model and domain. $n$ = number of resolved predictions. NBA excluded (see Web Appendix). Commod. = commodity futures.**

| Domain | Model | $n$ | H@50% | H@80% | H@90% | ECE |
|---|---|---|---|---|---|---|
| Stocks | Claude | 20 | 50% | 85% | 90% | 1.7% |
| | Gemini | 20 | 25% | 70% | 75% | 16.7% |
| | GPT-4 | 20 | 15% | 55% | 65% | 28.3% |
| Commod. | Claude | 5 | 20% | 40% | 40% | 40.0% |
| | Gemini | 5 | 20% | 40% | 40% | 40.0% |
| | GPT-4 | 5 | 20% | 40% | 40% | 40.0% |
| Crypto | Claude | 5 | 80% | 100% | 100% | 20.0% |
| | Gemini | 5 | 20% | 80% | 100% | 13.3% |
| | GPT-4 | 5 | 20% | 100% | 100% | 20.0% |
| Forex | Claude | 8 | 38% | 75% | 75% | 10.8% |
| | Gemini | 8 | 50% | 63% | 75% | 10.8% |
| | GPT-4 | 8 | 25% | 25% | 38% | 44.2% |
| Weather | Claude | 30 | 73% | 87% | 90% | 10.0% |
| | Gemini | 30 | 67% | 83% | 87% | 7.8% |
| | GPT-4 | 30 | 47% | 77% | 80% | 5.5% |

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**Meta-knowledge ratios** Table 5 presents the Soll–Klayman $\mu$ for each model-domain combination. All models show $\mu = 0.25$ on commodities—stated uncertainty is only one-quarter of what actual errors require, the most extreme overconfidence observed. At the other extreme, Claude's crypto $\mu = 14.4$ indicates CIs 14 times wider than necessary. GPT-4's high forex ECE (44.2%) is not fully captured by its $\mu$ (1.29), because $\mu$ cannot distinguish misanchored point estimates from miscalibrated CI widths; the two metrics must be examined jointly to diagnose the full failure mode.

**Table 5. Study 3: Meta-knowledge ratio $\mu$ = MEAD/MAD by model and domain. $\mu < 1$ = overconfidence; $\mu > 1$ = underconfidence.**

| Domain | Claude | Gemini | GPT-4 |
|---|---|---|---|
| Stocks | 0.81 | 0.55 | 0.41 |
| Commodities | 0.25 | 0.25 | 0.25 |
| Crypto | 14.40 | 1.35 | 1.84 |
| Forex | 2.68 | 1.67 | 1.29 |
| Weather | 1.28 | 1.04 | 0.85 |
| Overall | 1.23 | 0.99 | 0.81 |

***Discussion.*** The domain-confidence inversion is the central finding of Study 3: LLMs are most overconfident in structured financial markets where actual predictability is low, and most underconfident in speculative markets where they appear to apply large epistemic-humility priors. This parallels findings from the human expert literature—**(author?)** (12) showed that perceived predictability often exceeds actual predictability in information-rich, structured domains—and suggests that LLMs may inherit domain-specific miscalibration from training corpora dense in equity research and commodity price forecasts. The uniform $\mu = 0.25$ across all three models on commodities is particularly striking: this uniformity points to a shared domain prior in training data rather than a model-specific artifact. The crypto inversion (Claude $\mu = 14.4$) implies that models apply a "cryptocurrency is unpredictable" narrative that dramatically overstates typical 24-hour crypto volatility. The Study 3 aggregate model ranking replicates Studies 1 and 2 (Claude best, GPT-4 worst), but aggregate comparisons obscure the domain-crossing inversions that are the study's primary contribution.

## 2. General Discussion

The most consistent finding across all three studies is that frontier LLMs—particularly GPT-4—are substantially overconfident in their confidence intervals for quantitative forecasting. GPT-4's ECE ranged from 18.9% (Study 3, favorable domain mix) to 43.7% (Study 1, equity-only), with a cross-study mean exceeding 35%. Even Claude, the best-calibrated model in every study, achieved only 19.7% ECE in Study 1. This pattern of overconfidence was robust to domain, horizon, and sample size, confirming it as a systematic property of these models rather than an artifact of any single design.

These ECE values are comparable to the most severe human expert overconfidence on record. **(author?)** (6) found that CFOs' 80% intervals for annual equity returns achieved only 36–38% empirical coverage—roughly equivalent to our Study 1 Claude result (58% at 80%). GPT-4's Study 1 performance (31% at the 80% level) is dramatically worse, suggesting that some frontier LLMs may exhibit super-human overconfidence in certain forecasting contexts.

**The horizon effect** Study 2 reveals that prediction horizon strongly moderates miscalibration, with a divergent model-specific pattern. The degradation of Claude and GPT-4 is consistent with CI widths anchored to short-horizon volatility—a failure mode that mirrors human behavior at longer horizons (12). Gemini's near-perfect calibration at both 1 day and 22 days suggests its CI widths scale approximately proportionally to realized volatility at these extremes, though the mechanism underlying this unusual profile is not yet understood.

**The domain-confidence inversion** Study 3's central finding—that all models overstate certainty in commodity and equity markets while overstating uncertainty in cryptocurrency and forex—is inconsistent with a single domain-agnostic miscalibration mechanism. The most plausible interpretation is that LLMs apply domain-level heuristics inherited from training corpora: equity research and commodity analysis provide many narrow price forecasts, encoding an implicit prior that these markets are predictable, while cryptocurrency and forex commentary may emphasize unpredictability. The uniform $\mu = 0.25$ on commodities across all three models supports a shared training-data origin rather than a model-specific artifact.

**Model differences** GPT-4 was consistently the most overconfident model. Its Study 1 ECE (43.7%) is nearly double Claude's (19.7%) in the same task, and its Study 2 ECE at 22 days (49.9%) approaches the theoretical maximum for three-level calibration. Claude performed best in every study at the aggregate level but exhibits severe directional miscalibration in cryptocurrency ($\mu = 14.4$)—the largest positive outlier in the meta-knowledge analysis. This shows that even the best-calibrated model can be dramatically miscalibrated in specific domains. GPT-4's forex failure (ECE = 44.2%) is the worst domain-model result in Study 3 and reflects a compound failure of point-estimate anchoring and CI-width miscalibration that the MEAD/MAD ratio alone cannot fully capture; both metrics are required for complete diagnosis.

**Limitations** Several limitations constrain our conclusions. First, Studies 1 and 3 were each conducted on a single day, and Study 3's domain-specific results may partly reflect idiosyncratic market conditions on March 25, 2026. Study 2's backtest design mitigates this concern for equities but introduces potential look-ahead bias if models' training data extends to the backtest period. Second, statistical precision is limited in small Study 3 domains ($n = 5$ for commodities, $n = 5$ for crypto, $n = 8$ for forex). Third, the Claude model version differed between Studies 1–2 (claude-sonnet-4-20250514) and Study 3 (claude-sonnet-4-6); cross-study comparisons for Claude should be interpreted with this caveat. Fourth, NBA outcomes were excluded entirely from Study 3 due to ESPN API resolution failures. Fifth, models were evaluated without chain-of-thought or explicit calibration prompting; whether such interventions produce genuine calibration improvement is an important open question.

**Future directions** Future work should: (i) collect predictions daily over extended periods to separate stable calibration properties from single-day conditions; (ii) extend the domain set to include earnings forecasts, macroeconomic indicators, and prediction market prices; (iii) evaluate calibration-specific prompting and post-hoc recalibration (conformal prediction, temperature scaling); and (iv) use larger $n$ within Study 3 domains to enable robust domain-level inference.

## 3. Conclusion

We report three preregistered studies examining whether frontier LLMs exhibit systematic overconfidence in quantitative interval estimates, and how miscalibration varies with prediction horizon and domain. All three models were substantially overconfident across all three studies, with GPT-4 consistently the worst (cross-study ECE 19–44%) and Claude the best. Study 2 reveals that Claude and GPT-4 become increasingly miscalibrated as horizon grows from 1 to 22 days, while Gemini maintains near-target coverage at both ends of the tested range. Study 3 reveals a domain-confidence inversion: all models overstate certainty in commodity and equity markets while overstating uncertainty in cryptocurrency and forex—a pattern consistent with domain-specific miscalibration inherited from training corpora.

These results have direct practical implications. Users who elicit confidence intervals from LLMs for financial decision support should not assume that stated uncertainty reflects calibrated epistemic states. The pattern of miscalibration is most severe in precisely the domains (commodities, equities) and at precisely the horizons (multi-week) where overconfident forecasting carries the most severe consequences. Global calibration benchmarks do not transfer across domains or horizons; domain-stratified and horizon-stratified evaluation is necessary to characterize model-specific calibration risk. More broadly, this study demonstrates that the domain-specificity and horizon-sensitivity of human overconfidence—developed over four decades of research on human judges—apply with striking regularity to large language models trained on those same human outputs.

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1. M Alpert, H Raiffa, A progress report on the training of probability assessors in *Judgment Under Uncertainty: Heuristics and Biases*, eds. D Kahneman, P Slovic, A Tversky. (Cambridge University Press, Cambridge, UK), pp. 294–305 (1982).
2. S Lichtenstein, B Fischhoff, LD Phillips, Calibration of probabilities: The state of the art to 1980 in *Judgment Under Uncertainty: Heuristics and Biases*, eds. D Kahneman, P Slovic, A Tversky. (Cambridge University Press), pp. 306–334 (1982).
3. J Klayman, JB Soll, C Gonzalez-Vallejo, S Barlas, Overconfidence: It depends on how, what, and whom you ask. *Organizational Behavior and Human Decision Processes* **79**, 216–247 (1999).
4. JB Soll, J Klayman, Overconfidence in interval estimates. *Journal of Experimental Psychology: Learning, Memory, and Cognition* **30**, 299–314 (2004).
5. JJJ Christensen-Szalanski, JB Bushyhead, Physicians' use of probabilistic information in a real clinical setting. *Journal of Experimental Psychology: Human Perception and Performance* **7**, 928–935 (1981).
6. I Ben-David, JR Graham, CR Harvey, Managerial miscalibration. *Quarterly Journal of Economics* **128**, 1547–1584 (2013).
7. B Flyvbjerg, MK Holm, S Buhl, Underestimating costs in public works projects: Error or lie? *Journal of the American Planning Association* **68**, 279–295 (2002).
8. D Lovallo, D Kahneman, Delusions of success: How optimism undermines executives' decisions. *Harvard Business Review* **81**, 56–63 (2003).
9. S Kadavath, T Conerly, A Askell, , et al., Language models (mostly) know what they know (Preprint, arXiv:2207.05221) (2022).
10. S Lin, J Hilton, O Evans, Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research* **2022**, 1–24 (2022).
11. M Xiong, Z Hu, X Lu, , et al., Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs (Preprint, arXiv:2306.13063) (2024).
12. D Griffin, A Tversky, The weighing of evidence and the determinants of confidence. *Cognitive Psychology* **24**, 411–435 (1992).

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